

KYAW SWAR HEIN

SOFTWARE ENGINEER



CONTACT

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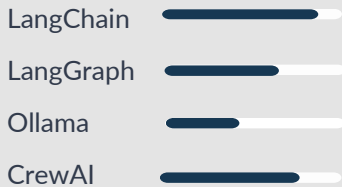
<https://ksh-space-portfolio-next.vercel.app/>

<https://github.com/Kyawswar-hein>

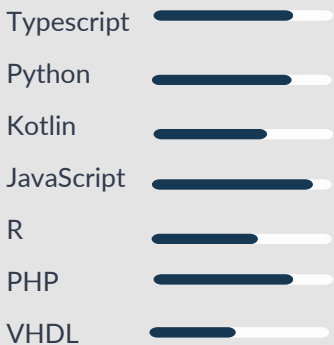
Tan Street, Insein Road, Hlaing Township, Yangon

SKILLS AND EXPERTISE

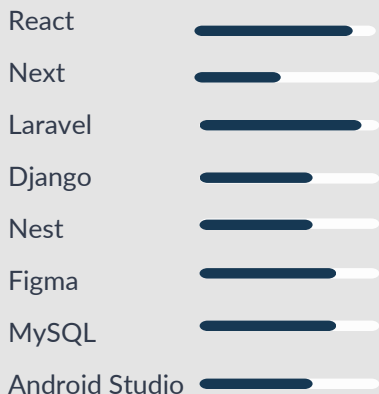
LLM & RAG



PROGRAMMING LANGUAGES



FRAMEWORKS AND TOOLS



PROFILE

"AI Engineer specializing in Agentic workflows, RAG pipelines, and full-stack integration. Proven track record building production-ready AI tools (n8n/LLMs) and robust backends (Spring Boot/Next.js). Passionate about automating operational workflows and deploying real-world AI systems."



EDUCATION

Myanmar Institute of Information Technology 2017 - 2026

- B.E. (Hons) | Electrical Communication Engineering

Brainworks International School (IGCSE) 2010 - 2015

- Secondary School | accredited by Cambridge University



WORK EXPERIENCE

DIR Ace Technology Company Ltd 2025 DEC - 2026 MAR

Fullstack Developer Intern

- Developed a Full-Stack Review Count Management System using Java Spring Boot and PostgreSQL to automate the tracking of project quality metrics (DQI, CQI, TQI).
- Architected a Role-Based Access Control (RBAC) system using Spring Security, implementing custom filters and success handlers for Admin, Editor, and Viewer roles.
- Data Import Module using Apache POI, enabling bulk processing of Excel datasets with transactional integrity and validation logic.
- Designed an Interactive Analytics Dashboard using Chart.js and Thymeleaf, providing real-time visual comparisons of "Base Week" vs. "Compare Week" performance.

Star88 Trading Company Ltd 2022 MAY - 2022 NOV

Bridge It Support (Project Based)

- Set up different devices, Barcode Scanners, printers for POS (point of sales) and ERP (enterprise resource planning features) of Retail and Wholesale Mart.
- Acted as bridge between the client company and the two software companies.
- Data cleaning, processing for over twenty thousand different items with price setting and barcode management then integrated into ERP software.

CLOUD SERVICES

- AWS (lightsail)
- Google Cloud
- Azure
- Firebase

LANGUAGES

- Burmese (Native)
- English (Business level)
- Japanese (Learning)

HOBBIES & INTERESTS

- Solving puzzles
- Watching Sci-Fi movies
- Listening to music
- Graphic design

FAVORITE QUOTATION

"Learning gives you the tools, but creation is what sharpens them."



PROJECT EXPERIENCES

Local AI Database Assistant & PostgreSQL Integration N8N

(Docker, n8n, PostgreSQL, OpenAI API / Large Language Models (LLMs))

- Designed a dual-agent AI system using n8n to translate natural language user queries into optimized, precise PostgreSQL queries.
- Implemented strict safety protocols, including hardcoded query limits (default 10, max 50) and error-prevention routing, mitigating system crashes and preventing unauthorized massive data retrieval.
- Designed a modular, commercial-ready workflow template capable of dynamically adapting to any PostgreSQL schema by separating user intent from SQL generation.
- Optimized system performance by leveraging a sub-workflow routing architecture, ensuring safe, scalable, and user-friendly data interaction for non-technical business teams.

Weapon Detection & Classification System

(TensorFlow, Keras, Scikit-Learn, NumPy, Matplotlib & Seaborn)

- Developed an end-to-end computer vision pipeline to identify and classify security threats (Knife, Longgun, Shotgun).
- Performed a comparative analysis between VGG16 and InceptionResNetV2 architectures using Transfer Learning to optimize for classification accuracy and model depth.
- Implemented InceptionResNetV2 with custom global average pooling and dropout layers to handle complex feature extraction from 299x299 image inputs.
- Utilized ImageDataGenerator for real-time data augmentation (rotation, zooming, and horizontal flipping) to improve model generalization and robustness against varied camera angles.
- Evaluated performance using Confusion Matrices and Seaborn visualization to identify and reduce false-positive rates between similar weapon categories.

Music Genre Classification - Machine Learning Webapp

(Django, Scikit-learn, Librosa)

- Engineered a machine learning pipeline to classify music genres by extracting audio features from .wav files using the Librosa library.
- Trained a Random Forest Classifier model on the extracted features and serialized it for production use.
- Built and deployed a user-friendly web interface with Django that allows users to upload audio files and receive instant genre predictions.



Student Management System -SMS

(Django, Tailwind, MySQL, SQLite)

- Designed and implemented a relational database schema in MySQL to efficiently manage student, course, and academic records.
- Developed a role-based web application with distinct user interfaces and permissions for Admins, Faculty, and Students.
- Automated key academic processes, including user management, course registration, and grade reporting, with custom logic for complex student requirements.

E-Grocer - Android Application

(Kotlin, Glide, Firebase)

- Developed a native Android e-commerce app in Kotlin, featuring a complete user flow from product browsing to checkout.
- Implemented the MVVM architecture, using Retrofit for API communication and the Room database for offline data caching.
- Integrated Firebase for secure user authentication, cloud data storage (Firestore), and media management (Cloud Storage).

3D Space-Themed Portfolio Website

(Next.js, Three.js, Framer Motion, TypeScript)

- Developed a visually immersive portfolio by integrating Three.js to render interactive 3D space-themed models and backgrounds.
- Engineered a dynamic and fluid user experience using Framer Motion for complex animations and smooth page transitions across all components.
- Built a fully responsive, multi-section website with Next.js 13 and TypeScript, and deployed it on Vercel for optimized performance and availability.

ABOUT ME

Passionate about building with technology, I see challenges as opportunities for growth. I am a dedicated learner, ready to apply my problem-solving skills and contribute to innovative solutions in a professional environment.